Autonomy calculation :

Given the components in my system, a Nucleo-32 L011K4 board, a MQ-4 sensor and a SIM7080G communication module, let's calculate the total current consumption and estimate the battery life of the whole system. We assume that the system captures and transmits data during one minute each hour.

Current consumption of components :

1. STM32L011K4 on Nucleo-32 board:
   - Deep sleep : 0.34 µA
   - Active mode : 5.7 mA (assuming 32 MHz clock speed)

2. MQ-4 Sensor:
   - Heating consumption = 750 mW
   - Typical consumption: 750 mW / 5V = 150 mA

3. SIM7080G NB-IoT Module:
   - Power saving mode: 3 µA
   - Active mode : 200 mA (assuming peak current during transmission)

Time in each mode :

Assume the system transmits data once every hour:
- Deep sleep: 59 minutes
- Active mode (STM32, MQ-4 sensor and SIM7080G transmission): 1 minute

Average current consumption :

1. STM32L011K4:
   - Deep sleep: 0.34 µA * 59 / 60 = 0.33 µA
   - Active: 5.7 mA * 1 / 60 = 95 µA

2. MQ-4 Sensor:
   - Active (for 1 minute/hour): 150mA * 1 / 60 = 2.5 mA

3. SIM7080G:
   - Sleep mode: 3 µA * 59 / 60 = 2.95 µA
   - Active transmission: 200 mA * 1 / 60 = 3.33 mA

Total average current consumption:

0.33 µA + 95 µA + 2.5 mA + 2.95 µA + 3.33 mA = 5.93 mA

Battery life calculation
Using two 1800mAh batteries in parallel (total 3600mAh):

3600 mAh / 5.93 mA = 605 hours = 25 days

References for STM32L011K4 :

Table 29. Typical and maximum current consumptions in Stop mode

Symbol	Parameter	Conditions	Typ	Max(1)	Unit
I_{DD} (Stop)	Supply current in Stop mode	T_A = -40°C to 25°C	0.34	0.99	µA
		T_A = 55°C	0.43	1.9	
		T_A= 85°C	0.94	4.2	
		T_A = 105°C	2.0	9	
		T_A = 125°C	4.9	19	

1. Guaranteed by characterization results at 125 °C, not tested in production, unless otherwise specified.

Table 27. Current consumption in Low-power Run mode

Symbol	Parameter	Conditions			Typ	Max(1)	Unit
I_{DD} (LP Run)	Supply current in Low-power run mode	All peripherals OFF, code executed from RAM, Flash switched OFF, V_{DD} from 1.65 V to 3.6 V	MSI clock, 65 kHz f_{HCLK} = 32 kHz	T_A = -40 °C to 25 °C	5.7	8.1	µA
				T_A = 85 °C	6.5	9	
				T_A = 105 °C	8	13	
				T_A = 125 °C	11.5	22	
			MSI clock, 65 kHz f_{HCLK} = 65 kHz	T_A =-40 °C to 25 °C	8.7	11	
				T_A = 85 °C	9.5	12	
				T_A = 105 °C	11	15	
				T_A = 125 °C	15	24	
			MSI clock, 131 kHz f_{HCLK} = 131 kHz	T_A = -40 °C to 25 °C	17	19	
				T_A = 55 °C	17	19.5	
				T_A = 85 °C	17.5	20	
				T_A = 105 °C	19	22	
				T_A = 125 °C	22.5	31	
		All peripherals OFF, code executed from Flash, V_{DD} from 1.65 V to 3.6 V	MSI clock, 65 kHz f_{HCLK} = 32 kHz	T_A = -40 °C to 25 °C	18	22	
				T_A = 85 °C	20	24	
				T_A = 105 °C	22	27	
				T_A = 125 °C	26.5	37	
			MSI clock, 65 kHz f_{HCLK} = 65 kHz	T_A = -40 °C to 25 °C	22	25	
				T_A = 85 °C	24	27	
				T_A = 105 °C	26	30	
				T_A = 125 °C	30.5	39	
			MSI clock, 131 kHz f_{HCLK} = 131 kHz	T_A = -40 °C to 25 °C	32	34	
				T_A = 55 °C	32.5	35	
				T_A = 85 °C	34	37	
				T_A = 105 °C	36	39	
				T_A = 125 °C	40	47	

1. Guaranteed by characterization results at 125 °C, not tested in production, unless otherwise specified.

References for MQ4 Sensor :

A. Standard work condition

Symbol	Parameter name	Technical condition	Remarks
Vc	Circuit voltage	5V±0.1	AC OR DC
V_H	Heating voltage	5V±0.1	ACOR DC
P_L	Load resistance	20KΩ	
R_H	Heater resistance	33Ω ±5%	Room Tem
P_H	Heating consumption	less than 750mw	

References for SIM7080G NB-IoT Module :

Table 31: Current consumption on VBAT Pins (VBAT=3.8V)

GNSS	
GNSS supply current (AT+CFUN=0,without USB connection)	Tracking, typical: 52mA
Sleep/idle mode	
LTE supply current (GNSS off, without USB connection)	Sleep mode Typical: 1.2mA Idle mode Typical: 14mA
Power Saving Mode	
PSM supply current	PSM mode Typical: 3uA
e-DRX	
e-DRX mode supply current (Tested in sleep mode)	@PTW=40.96s; eDRX=81.92s; DRX=2.56s, Typical: 1.4mA
	@PTW=25.6s; eDRX=163.84s; DRX=2.56s, Typical: 0.59mA
LTE Cat-M1 Voice Call	
TBD	TBD
TBD	TBD

LTE Cat-NB1/NB2 Data Transmission (15KHz single tone)	LTE-FDD B13	@21dbm Typical: 142mA @10dbm Typical: 82mA @0dbm Typical: 54mA
	LTE-FDD B18	@21dbm Typical: 144mA @10dbm Typical: 83mA @0dbm Typical: 54mA
	LTE-FDD B19	@21dbm Typical: 145mA @10dbm Typical: 83mA @0dbm Typical: 54mA
	LTE-FDD B20	@21dbm Typical: 146mA @10dbm Typical: 83mA @0dbm Typical: 54mA

Typical consumption for NB IoT transmission